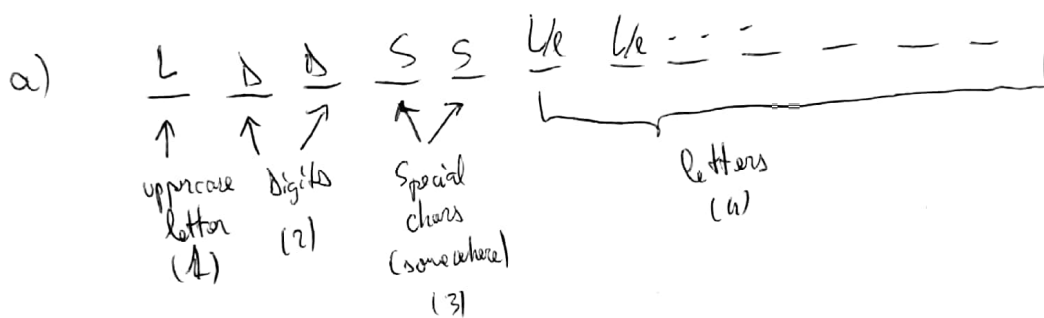


Problem 3

- 12 chars.
- pos 1 $\rightarrow$ uppercase
- pos 2,3 distinct digits
- 2 special chars
- 7 letters.



- (1) Letter upper $\rightarrow$ 26 possibilities
 (2) 2 Distinct digits $\rightarrow$ 10 $\cdot$ 9 possibilities
 (3) 2 Special chars $\rightarrow$ 7 $\cdot$ 6 possibilities

Those have to be placed somewhere: C_9^2 positions = $\frac{9!}{4! \cdot 2!} = \frac{9 \cdot 8}{2} = 9 \cdot 4$

(4) 7 letters : 52 letters possible $\Rightarrow$ 52^7 possibilities

$$\begin{aligned} \Rightarrow \text{Total number of keys} &= 26 \cdot 10 \cdot 9 \cdot 7 \cdot 6 \cdot C_9^2 \cdot 52^7 \\ &= 3538080 \cdot 52^7 \\ &= 3637399929280266246 \\ &\approx 3.6 \cdot 10^{18} \end{aligned}$$

With a poor approximation $10^3 \approx 2^{10}$
 $\Rightarrow 3.6 \cdot 2^{60}$

- b) No because the keyspace is too big.
 Too much to calculate. A lifetime is not enough.
 2^{55} universe? something like that.
 $\nearrow$ universe creation.